

[HEPro AI+] Activity 1: Dataset Design & Student Profiling

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Project: Dedicated Mentoring System for Students

1. Project Overview

This document outlines the design of the `students.csv` dataset. The dataset serves as the primary input for representing student profiles through multi-dimensional feature vectors across Academic, Wellness, Productivity, and Career domains.

2. Data Dictionary

The following table serves as the metadata reference for the student feature vectors.

Column Name	Description	Range
<code>student_id</code>	Unique identifier for each student	S1001 - S1060
<code>age</code>	Age of the student	18 – 25
<code>program</code>	Degree stream (B.Tech, MBA, M.Sc, B.A)	Categorical
<code>semester</code>	Current academic semester	1 – 8

gpa	Grade Point Average	0.0 – 10.0
attendance	Percentage of classes attended	0 – 100%
assignments_completion	% of coursework completed on time	0 – 100%
stress_level	Self-reported stress (10 = Highest)	1 – 10
sleep_hours	Average hours of sleep per night	0 – 12
mental_wellbeing	Self-reported wellbeing (10 = Best)	1 – 10
productivity_score	Personal efficiency rating	1 – 10
distractions	Frequency of focus interruptions	1 – 10
career_clarity	Clarity of professional goals	1 – 10
skill_readiness	Alignment with job market skills	1 – 10

engagement_score	Platform/Community interaction	0 – 100
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3. Data Insights on Real Student Behavior

By analyzing the dataset of 60 students, we can identify key behavioral trends that mirror real world challenges in higher education. These insights allow the system to support proactive mentoring.

Insight 1: Burnout Correlation

- **Data Evidence:** Students **S1001 to S1005** exhibit a high stress_level of **9** and a very low productivity_score of **2**.
- **Real student Behavior :** This identifies students struggling with a high mental load. Even when sleep hours are sufficient, the low mental_wellbeing (score of 3) indicates that psychological pressure is significantly hindering their ability to remain productive.

Insight 2: High Effort, Low Output

- **Data Evidence:** Students **S1011 to S1015** maintain a high engagement_score of **92** and perfect attendance of **95**, yet their gpa remains low at **5.5**.
- **Real student Behavior :** This represents students who are highly committed and disciplined but are failing to grasp the core material. They are physically present and active but lack effective study techniques or foundational knowledge.

Insight 3: Academic Success vs Career Uncertainty

- **Data Evidence:** Students **S1021 to S1024** possess a high gpa of **9.4** but have a very low career_clarity score of **2**.
- **Real student Behavior :** This highlights High Potential students who are academically excellent but lack a post-graduation roadmap. They are succeeding in exams but remain unprepared for the transition to the professional world.

